

THE PISTOL SHRIMP CLAW

A Biologically Encoded, Multi-Scale Energy-Routing System

Geometry, cavitation, structural wave management, and the engineering problem of surviving repeated extreme events

ENERGY TRANSFORMATION AND SPATIAL EXPORT



Design objective: route the extreme event into the fluid while limiting damaging return paths into the organism.

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Abstract

The snapping shrimp claw should not be understood merely as a biological mechanism that produces a fast jet or an intense cavitation collapse. Its more consequential engineering achievement lies in producing that event repeatedly while preserving the generating apparatus and the organism. The enlarged snapper claw combines a latch-mediated hinge, rotationally accelerated dactyl, plunger-socket displacement chamber, restricted outlet, vortex-forming pulse, differentiated cuticle, specialized surface morphology, and organism-level shock protection. These features form a coupled fluid-solid architecture in which energy undergoes a controlled sequence of transformations and spatial redistributions.

muscle work -> elastic strain -> rotational kinetic energy -> fluid displacement -> jet momentum -> vorticity -> cavitation -> shock, heat, and acoustic radiation

The design problem therefore concerns not only peak output but directionality, impedance, fatigue resistance, thermal and mechanical isolation, recovery, and repeatability across duty cycles. The claw can be interpreted as a multi-scale energy-routing instrument whose external form and internal material geometry jointly govern where energy accumulates, transforms, leaves the structure, and returns. This interpretation is compared with tokamaks, stellarators, turbine engines, rocket chambers, and atmospheric-entry systems, where generation of an extreme state is often less difficult than surviving its heat, particle, pressure, or stress flux over useful operating life.

Scope and evidentiary discipline

This document distinguishes three levels of statement: (1) observations directly reported for snapping shrimp, including latch-mediated closure, plunger-socket jet generation, vortex cavitation, species-dependent kinematics, and orbital-hood shock damping; (2) standard governing equations used as idealized models of those processes; and (3) engineering interpretations and testable hypotheses, especially the description of the cuticle as an impedance-shaped mechanical waveguide. The term waveguide is used operationally for geometry or material architecture that controls propagation, reflection, dispersion, attenuation, and directionality. It does not imply that every claw layer has a demonstrated phononic band gap.

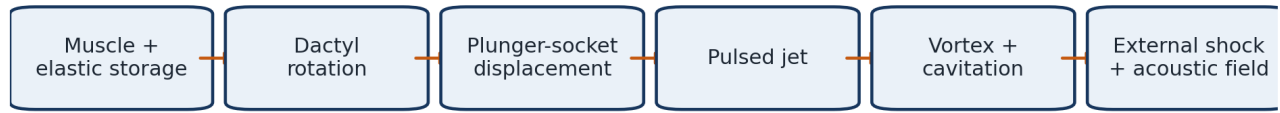
1. The engineering question: output versus survivability

A single violent cavitation event is not the complete achievement. The demanding problem is to reload, fire, recover, and repeat without destructive solid impact, rapid erosion, misalignment, neural injury, or prohibitive metabolic cost. A useful performance measure must therefore include both useful output and lifetime.

$$Pi_{surv} = (E_{useful} * N_{cycles}) / (E_{input} * D_{accum})$$

Here E_{useful} denotes energy transferred into the external fluid event, N_{cycles} the functional number of snaps, E_{input} the biological work required to load and actuate the mechanism, and D_{accum} a measure of accumulated damage. A design with lower peak output but far greater cycle life may produce more useful lifetime work.

ENERGY TRANSFORMATION AND SPATIAL EXPORT



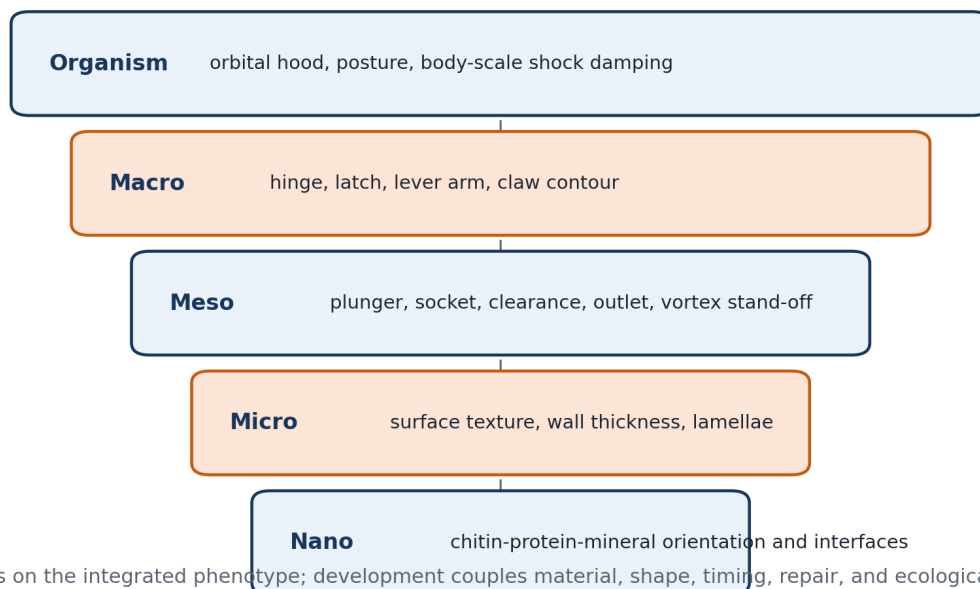
Design objective: route the extreme event into the fluid while limiting damaging return paths into the organism.

Figure 1. The claw exports stored biological energy into a fluid event through a sequenced series of transformations. The design problem includes the outgoing path and the returning mechanical and acoustic paths.

2. Developmental geometry and distributed optimization

DNA does not encode a claw as a list of final Cartesian coordinates. Genetic and epigenetic regulatory systems control growth, cell differentiation, secretion, mineralization, tissue boundaries, orientation fields, molting, and remodeling. These processes construct a hierarchy in which molecular organization, lamellar architecture, local wall thickness, surface texture, hinge form, and whole-claw kinematics arise through one developmental program.

BIOLOGICALLY ENCODED GEOMETRY ACROSS SCALES



Selection acts on the integrated phenotype; development couples material, shape, timing, repair, and ecological performance.

Figure 2. The function spans organismal, macroscopic, mesoscopic, microscopic, and nanoscopic scales. Each scale changes the effective loading seen by adjacent scales.

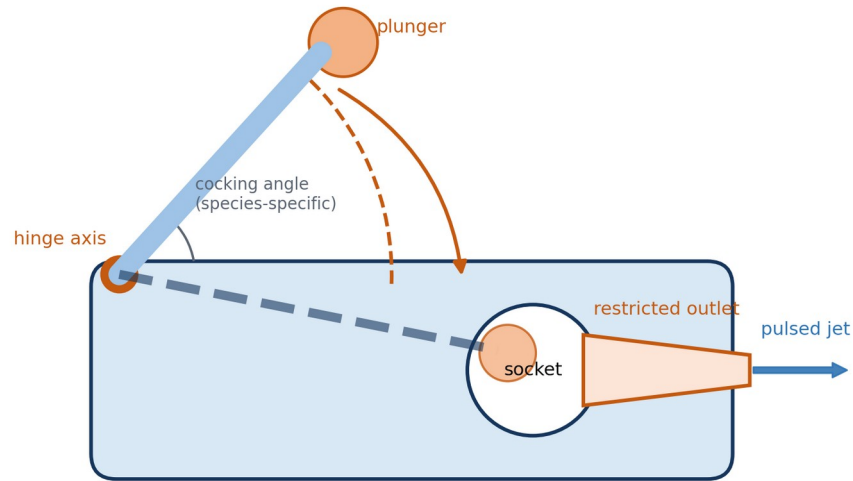
Biological evolution is not a conscious optimizer and need not locate a global mathematical optimum. It performs a historical search over heritable developmental variation. Natural selection filters integrated phenotypes according to organismal performance within an environment. A conceptual fitness functional can be written as:

$$F(g) = w_1 J(g) - w_2 \sigma_{\max}(g) - w_3 D_f(g) - w_4 C_m(g) - w_5 T_r(g) + w_6 A(g)$$

The developmental parameter set g may simultaneously affect jet performance J , maximum stress $\sigma_{\max}$, fatigue accumulation D_f , material and metabolic cost C_m , reload time T_r , and ecological advantage A . No one term must be individually maximized. The viable phenotype represents a coupled compromise constrained by ancestry, seawater properties, available biomaterials, molting, and ecological use.

3. The snapper claw as a tuned three-dimensional instrument

The enlarged claw contains a movable dactyl and a fixed propodus. The dactyl bears a plunger that enters a corresponding socket in the propodus. Rapid closure displaces water through a restricted opening and produces a pulsed jet. High-speed imaging, CT-derived models, transparent enlarged models, and numerical simulations support the plunger-socket displacement mechanism and the formation of a leading vortex [1-4].



The terminal trajectory must align the plunger with the socket while preserving clearance and directing displaced water through the outlet.

Figure 3. Functional geometry of the cocked and closing snapper claw. Exact angles and proportions differ among species; the essential relationship is the constrained terminal trajectory from hinge rotation to plunger insertion and outlet-directed displacement.

The system behaves more like a spring-driven positive-displacement pulse generator than like a pair of impacting jaws. The hinge constrains trajectory; the latch permits preload; the lever arm converts angular motion into plunger speed; the socket confines water; and the outlet conditions the pulse and its shear layer. The terminal geometry must preserve clearance while directing displaced water outward rather than converting the stored energy into destructive solid-on-solid impact.

3.1 Hinge, latch, and torque reversal

Comparative work identifies cocking slip joints and torque-reversal joints as key evolutionary innovations in snapping shrimp [2]. The muscle line of action and effective moment arm change with joint angle. The torque can be represented as:

$$\tau(\theta) = F_m(\theta) * r_m(\theta)$$

The rotational dynamics are approximated by:

$$I(\theta) * \ddot{\theta} = \tau_s(\theta) - \tau_h(\theta, \dot{\theta}) - \tau_f(\theta, \dot{\theta})$$

where I is effective rotational inertia, τ_s stored elastic or muscle-generated torque, τ_h hydrodynamic resistance, and τ_f joint friction and internal damping. Reported peak angular velocities vary substantially by species and method, with values in the approximate range of 10^3 to 4.5×10^3 rad/s in one multi-species study [1]. Closure times as short as roughly 0.36 ms have been reported in related performance studies [5]. These figures should not be treated as universal constants.

3.2 Trajectory precision

Speed alone does not guarantee efficient pulse formation. Let v_p be plunger-tip velocity and n_s the socket insertion axis. The useful axial component is:

$$v_{\text{parallel}} = v_p \cos(\alpha), \quad v_{\text{perpendicular}} = v_p \sin(\alpha)$$

$$v_{\text{perpendicular}} / v_{\text{parallel}} = \tan(\alpha)$$

Even modest misalignment α increases lateral fluid acceleration, wall loading, leakage, and the possibility of local contact. Precision therefore resides in the relation among hinge-axis location, angular excursion, lever arm, plunger curvature, socket axis, clearance, outlet contour, and release timing. A functional geometric parameter set can be written as:

$$G = \{\theta_c, r_p, R_p, R_s, L_s, A_o, \beta_o, \delta, \alpha\}$$

The output depends on the coupled vector G rather than any one dimension in isolation. This is the sense in which the claw resembles a highly tuned instrument: small changes in geometry can alter phase, alignment, leakage, pulse duration, vortex circulation, and self-loading.

4. The plunger-socket pair as a transient displacement nozzle

Let $V_c(t)$ denote the water-filled chamber volume. Its displacement rate is:

$$Q_d(t) = -dV_c/dt$$

For a short pulse, conservation of volume gives approximately:

$$Q_o(t) + Q_l(t) \approx -dV_c/dt$$

$$U_o(t) = Q_o(t)/A_o(t) \approx [-dV_c/dt - Q_l(t)] / A_o(t)$$

If x denotes insertion depth and $A_{p,\text{eff}}$ the effective plunger area, then:

$$-dV_c/dt \approx A_{p,\text{eff}}(x) * \dot{x}$$

$$U_o \approx [A_{p,\text{eff}}/A_o] * \dot{x} - Q_l/A_o$$

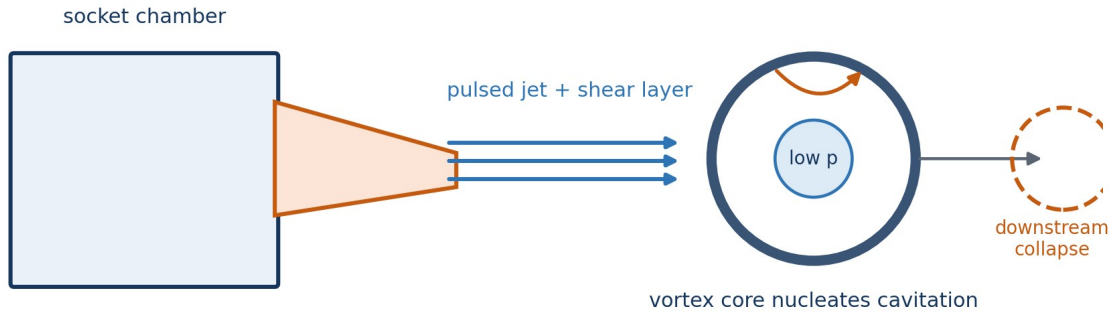
The area ratio $\Gamma_A = A_{p,\text{eff}}/A_o$ acts as a nominal velocity-amplification factor, but only within a coupled fluid-structure system. Making the outlet indefinitely smaller would increase hydraulic resistance, local pressure, sensitivity to fouling, viscous loss, and wall loading. The biological form therefore suggests tuning toward a survivable operating region rather than maximizing constriction alone.

5. Vortex formation, cavitation, and controlled stand-off

The emerging pulse enters nearly stationary seawater and produces a surrounding shear layer. The vorticity is $\omega = \text{curl}(u)$, and circulation is:

$$\Gamma = \int_C \mathbf{u} \cdot d\mathbf{l} = \int_S \boldsymbol{\omega} \cdot \mathbf{n} dS$$

Experiments with enlarged transparent claw models show that the pulse forms a leading vortex and that cavity evolution depends strongly on vortex dynamics [3]. Numerical and high-speed studies likewise locate the pressure reduction near the nozzle and in the vortex structure [1,4].



The geometric objective is not merely low pressure; it is controlled formation, stand-off, collapse location, and repeatability.

Figure 4. The outlet does more than accelerate water. It helps define the pulse, shear layer, vortex circulation, low-pressure core, and downstream location of cavitation collapse.

The cavitation number is:

$$\sigma = (p_{\infty} - p_v) / (0.5 \rho U^2)$$

Lower σ indicates greater susceptibility to cavitation. In a swirling region, radial equilibrium gives:

$$dp/dr = \rho * u_{\theta}^2 / r$$

$$p(r) = p(R) - \int_r^R \rho u_{\theta}(r_{\text{prime}})^2 / r_{\text{prime}} dr_{\text{prime}}$$

A sufficiently strong tangential velocity depresses core pressure. Cavitation becomes possible when the local pressure approaches vapor pressure, with nucleation also depending on dissolved gas, nuclei, temperature, surface condition, and pressure history.

5.1 Pulse duration and vortex loading

The system must tune not only peak speed but the entire pulse waveform. A dimensionless stroke ratio is:

$$\Lambda_s = L_j/D_o = [1/D_o] * \int_0^{T_j} U_o(t) dt$$

A pulse that is too short may deposit insufficient circulation. A pulse that is too long may generate trailing flow that no longer contributes efficiently to the leading vortex. Outlet diameter, lip shape, displaced volume, closure waveform, and pulse duration therefore form a coupled vortex-generation problem.

5.2 Bubble evolution

A Rayleigh-Plesset-type equation provides an idealized model of bubble radius $R(t)$:

$$\rho [R \ddot{R} + (3/2) \dot{R}^2] = p_B(t) - p_\infty(t) - 2 \gamma/R - 4 \mu \dot{R}/R$$

The observed cavity can be toroidal, advected, asymmetric, rebounding, and gas-containing, so a spherical model remains an approximation. An idealized collapse time is:

$$t_c \sim 0.915 R_{\max} \sqrt{\rho/(p_\infty - p_v)}$$

The collapse energy partitions among shock, acoustic radiation, fluid microjets, heat, rebound, and residual motion. The system does not need to maximize each channel; it needs to produce an ecologically useful external pressure event while controlling the structural cost of doing so.

5.3 Spatial export of the extreme event

The central geometric achievement is stand-off. By forming a jet and downstream vortex, the claw relocates much of the most violent fluid event away from the plunger-socket interface. Define a normalized collapse stand-off:

$$S_c = x_c / D_o$$

Too small a stand-off increases near-wall erosion and return loading. Too large a stand-off may reduce useful pressure at close range. The first directional requirement is therefore not simplistic focusing on a target. It is export from the generating material into a fluid structure that can carry and transform the energy externally.

6. Material architecture as an impedance-shaped wave path

The material is not a passive container added after the mechanism has been designed. Its spatial organization is part of the mechanism. Density, modulus, mineral concentration, porosity, hydration, fiber orientation, lamellar pitch, wall thickness, and curvature determine how stress waves propagate, reflect, refract, disperse, and attenuate.

For a longitudinal elastic wave, a local mechanical impedance can be approximated as:

$$Z_L = \rho_s c_L$$

At a sharp normal interface, the pressure-amplitude reflection coefficient is:

$$R_{\text{amp}} = (Z_2 - Z_1)/(Z_2 + Z_1)$$

$$R_{\text{energy}} = |(Z_2 - Z_1)/(Z_2 + Z_1)|^2$$

A graded transition spreads the impedance change through depth and can reduce a concentrated reflected pulse, although the exact result depends on wavelength, anisotropy, damping, geometry, and loading direction. A qualitative slowly varying condition is:

$$|(1/Z) dZ/dx| \ll k$$

where k is local wavenumber. The phrase waveguide therefore denotes the combined effect of boundaries and constitutive gradients on preferred stress-wave paths. It should not be confused with a claim that the entire claw behaves as a lossless one-dimensional guide.

6.1 Cuticle hierarchy and fracture routing

Published microscopy and mechanical studies describe differentiated cuticular regions, specialized surface morphology, and a sandwich-like plunger architecture with harder outer regions and a distinct internal core [1,6]. More generally, helicoidal or Bouligand architectures rotate reinforcement direction through thickness. An idealized orientation field is:

$$\mathbf{n}(z) = [\cos(\theta(z)), \sin(\theta(z)), 0], \quad \theta(z) = \theta_0 + q z, \quad P = 2 \pi/q$$

This produces a spatially varying anisotropic stiffness tensor $C(z)$. Cracks and stress waves encounter changing directions and properties, promoting crack twisting, deflection, branching, distributed strain, and frequency-dependent transmission. Direct phononic-bandgap measurements are well established for mantis-shrimp dactyl-club architecture, but should not be transferred automatically to pistol-shrimp cuticle. For the pistol shrimp, specific band gaps remain a species-, region-, hydration-, and frequency-dependent hypothesis requiring direct measurement.

6.2 Surface geometry

Cone-shaped micropapillae and finer textures have been reported on snapper-claw surfaces, with computational interpretations relating them to hydrodynamic resistance during rapid closure [1,6]. Drag can be represented as:

$$F_D = 0.5 \rho C_D A U^2$$

$$W_D = \int_{\theta_0}^{\theta_f} \tau_D(\theta, \dot{\theta}) d\theta$$

Because hydrodynamic force scales approximately with U^2 , modest changes in effective drag can matter during ultrafast motion. Surface features may also affect abrasion, wetting, nucleation, turbulence, reinforcement, and crack initiation. Multifunctionality is more plausible than a one-feature/one-function mapping.

7. Organism-level return-wave protection

The energy-routing system extends beyond the claw. Recent experiments and structural analyses indicate that snapping-shrimp orbital hoods damp self-generated shock waves and reduce short-term blast effects on sensitive tissues [7]. This provides direct support for a two-zone protection model: source management at the claw and return-wave management at the head and body.

- Source management: latch, hinge, plunger, socket, outlet, jet, and downstream cavitation.
- Return-wave management: propagation through surrounding water, body-scale coupling, hood deformation and damping, and reduced transmission to eyes and neural tissue.

The organism therefore solves not only the outward event but also the field that returns after the event has left the claw.

8. Duty-cycle-first efficiency

Ordinary energy efficiency is incomplete for a repeated extreme-event system:

$$\eta_E = E_{\text{external}} / E_{\text{stored}}$$

A lifetime measure includes repair, replacement, and accumulated damage:

$$\eta_{\text{life}} = [N E_{\text{external}}] / [N E_{\text{stored}} + E_{\text{repair}} + E_{\text{replacement}}]$$

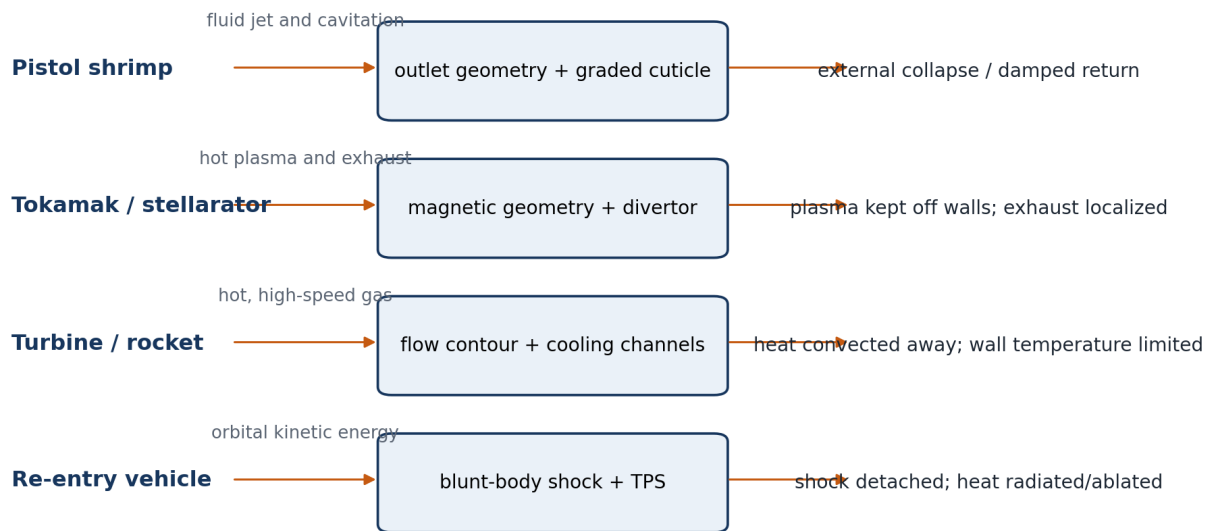
A simple fatigue accumulation model is:

$$D = \sum_j n_j / N_j$$

Biological cuticle complicates classical fatigue analysis through hydration, viscoelasticity, molting, remodeling, and repair. Yet the design principle remains: peak output cannot be assessed independently from the number of cycles over which alignment, material integrity, and organismal function persist.

9. Cross-domain comparison: the architecture of surviving generation

COMMON ARCHITECTURE: GENERATE - ROUTE - ISOLATE - SURVIVE - REPEAT



The analogies concern design logic, not identical physical mechanisms.

Figure 5. Four systems with different physics share a design logic: generate an extreme state, organize its path geometrically, isolate vulnerable material, route residual loads, and repeat.

9.1 Tokamaks and stellarators

Fusion devices can create extremely hot plasma; the decisive engineering problem is confining it, exhausting energy and particles, and preventing unacceptable damage to plasma-facing components. In ITER, the divertor vertical targets are designed for approximately 10 MW/m^2 steady heat flux and 20 MW/m^2 during slow transients [8]. These figures make duty cycle and thermal fatigue first-order design constraints, not secondary details.

In a tokamak, magnetic geometry organizes charged-particle trajectories and concentrates exhaust in the divertor, where specially selected armor, cooling channels, replaceable cassettes, and detachment strategies manage the load. In a stellarator, three-dimensional coil and magnetic-surface geometry becomes even more explicit. Wendelstein 7-X was numerically optimized to investigate steady-state-compatible stellarator confinement and long-pulse operation [9].

The analogy with the shrimp concerns architecture, not identical mechanism: field geometry reduces the burden before material strength must carry the residual load. The shrimp exports the event into a vortex and downstream cavity; the fusion device keeps the hottest plasma from directly contacting the wall while routing unavoidable exhaust toward designed interfaces.

9.2 Turbine engines

Modern gas turbines can expose components to gas temperatures beyond the uncooled capability of blade materials. Internal convection, impingement cooling, film cooling, coatings, and alloy

microstructure operate together. NASA describes film cooling as compressor air ducted through blade chambers and discharged through shaped holes to form protective cooling jets [10].

$$q'' = h_g (T_g - T_w)$$

$$\eta_f = (T_g - T_{aw}) / (T_g - T_c)$$

The blade survives not because one material passively tolerates the complete hot-gas environment, but because geometry continually channels coolant, modifies the near-wall temperature field, and transports heat away. The cooling-hole shape, internal passage layout, wall thickness, coating, and external aerodynamics form one thermal-fluid instrument.

9.3 Rocket chambers and nozzles

Rocket engines intensify the same problem. Regenerative cooling sends propellant through channels in the chamber and nozzle wall before injection. The channels remove heat, preheat propellant, and protect the structure; the nozzle then converts pressure and enthalpy into directed exhaust momentum.

$$h_0 = h + U^2/2$$

$$F = \dot{m} U_e + (p_e - p_a) A_e$$

The throat can experience exceptional heat flux and stress. Survivability depends on contour geometry, coolant distribution, injector uniformity, conductivity, wall thickness, coatings, fatigue, and transient startup and shutdown. A chamber that generates exceptional thrust once but burns through its throat has not solved the engineering problem.

9.4 Atmospheric entry

Atmospheric entry converts orbital kinetic energy into shocked, compressed, reacting gas surrounding the vehicle. A blunt body creates a detached bow shock and keeps the strongest shock away from the surface. Thermal-protection materials then manage the residual energy through insulation, radiation, conduction control, pyrolysis, ablation, and mass loss [11].

$$E_k = 0.5 m v^2$$

$$q_{in}'' = q_{cond}'' + q_{rad}'' + \dot{m}_a h_a + q_{stored}''$$

Ablation deliberately assigns material to carry energy away. This is not failure in the ordinary sense; it is a controlled energy path. As in the shrimp, shape and material must be co-designed around where the extreme state occurs and how much of it reaches the protected interior.

10. A general theory of survivable extreme-event systems

Across these examples, five functional layers recur:

- Energy storage or generation.
- Geometric conversion into an organized field, flow, or trajectory.
- Spatial separation of the extreme state from vulnerable components.
- Material routing, attenuation, cooling, or sacrificial removal of residual loads.
- Recovery, inspection, repair, replacement, or biological renewal enabling repetition.

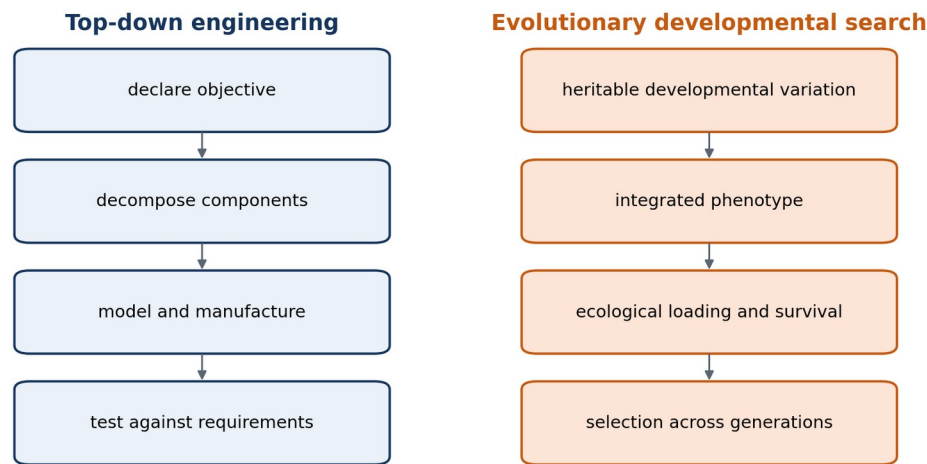
$$x(t + T) \sim x(t), \text{ subject to accumulated damage remaining below } D_{crit}$$

A general survivability functional may be written as:

$$S = [\eta_u P_u T_{life}] / \max_{(x,t)}[\sigma_{eq}/\sigma_{allow} + T/T_{allow} + q_{doubleprime}/q_{allow_doubleprime} + D/D_{crit}]$$

The terms vary by system, but the objective does not: maximize useful lifetime output subject to structural, thermal, biological, and energetic constraints.

11. Biological self-optimization and top-down engineering



Engineering can borrow the biological objective: optimize whole-life performance, interfaces, and duty cycle rather than peak output alone.

Figure 6. Evolutionary developmental search and top-down engineering use different search procedures, but both can be evaluated against whole-life system performance.

Biological evolutionary development	Conventional top-down engineering
Searches heritable developmental variation	Searches explicit models, dimensions, materials, and controls
Filters integrated phenotype in an environment	Decomposes requirements and components
Uses growth, self-assembly, molting, and repair	Uses controlled fabrication, assembly, inspection, and replacement
Often produces multifunctional graded structures	Often favors modular specialization and discrete interfaces
Optimizes reproductive viability under historical constraints	Optimizes declared objectives under design and manufacturing constraints
Cannot freely abandon ancestry or available biochemistry	Can introduce non-biological fields, materials, and manufacturing processes

The useful lesson is not that biology always outperforms engineering. It is that engineering objectives can be broadened toward biological survivability: optimize interfaces; treat material architecture as geometry; design the return path as carefully as the outgoing path; include startup, shutdown, recovery, and repair; and optimize lifetime useful work rather than peak output alone.

12. Testable engineering hypotheses

The wave-routing interpretation generates measurable hypotheses rather than functioning only as metaphor.

- Controlled stand-off: a pulsed-fluid device can improve source lifetime by shaping vortex formation so the dominant collapse occurs at a controlled $S_c = x_c/D_o$.
- Graded socket impedance: an axial or radial modulus gradient can reduce peak reflected stress relative to a uniform socket while retaining useful outlet momentum.
- Coupled geometric tuning: performance should be optimized over hinge, plunger, socket, clearance, outlet, and pulse parameters together rather than maximizing closure speed alone.
- Surface-mediated drag control: claw-inspired microtextures can be tested at relevant Reynolds numbers for drag, separation, wetting, and nucleation effects.
- External protective hoods: compliant or impedance-graded covers can reduce transmission of self-generated underwater shocks to sensitive equipment.
- Duty-cycle-first optimization: geometry maximizing $N_{cycles} E_{useful}/cycle$ may differ substantially from geometry maximizing one-shot cavitation intensity.

13. Measurements required for a predictive model

Domain	Required measurements
Geometry	3D hinge axis; open and closed angles; plunger/socket curvature; outlet area and contour; clearance versus insertion depth; wall thickness; surface texture.
Kinematics	$\theta(t)$, angular velocity and acceleration; latch timing; plunger trajectory; elastic deformation of dactyl and propodus.
Fluid field	Velocity, pressure, vorticity, vapor fraction, bubble topology, stand-off, acoustic pressure, rebound and cloud formation.
Material field	Mineral composition, chitin orientation, density, porosity, hydration, elastic moduli, damping, fracture toughness, conductivity, fatigue and wave-speed spectra.
Organismal response	Body pressure, orbital-hood deformation, acceleration near neural tissue, snap repetition rate, molt-to-molt damage and repair.

A complete model would couple multibody dynamics, unsteady multiphase flow, anisotropic viscoelastic solids, fracture and fatigue, acoustics, and organismal boundary conditions. The apparent precision arises not from one exact number but from the co-registration of many tolerances and timescales.

14. Final synthesis

The pistol shrimp claw represents a biological architecture in which developmental processes construct geometry across scales; latch and hinge geometry store and release energy; lever geometry controls acceleration; plunger and socket geometry convert motion into fluid displacement; outlet geometry shapes a pulse; the pulse organizes vorticity; the vortex creates a low-pressure core; cavitation relocates the most extreme event into the surrounding water; differentiated cuticle manages residual solid stress; and body-scale structures damp returning shock.

Its deepest engineering lesson does not concern how to produce the largest possible cavitation bubble. It concerns how to organize an extreme process so the structure that generates it does not have to endure the full event directly.

[where energy forms] -> [where it travels] -> [where it transforms] -> [where it leaves] -> [what remains protected]

Across snapping claws, fusion devices, turbine and rocket hot sections, and atmospheric-entry systems, generation alone never completes the design. A viable system must survive its own success over time. The pistol shrimp therefore offers a model of survivable multi-scale energy choreography: an exacting relation among developmental instruction, shape, timing, fluid properties, material impedance, structural hierarchy, recovery, and repeatable operation.

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Editorial and mathematical audit note

Equations in this document are governing or scaling relations, not claims of a complete closed-form model. Continuity, torque, radial equilibrium, Rayleigh-Plesset dynamics, impedance reflection, fatigue accumulation, heat transfer, thrust, and energy balance are dimensionally consistent in the forms presented. Symbols are defined locally. Approximation signs mark idealizations. Species-variable numerical values are reported as ranges or examples rather than universal specifications. Claims about direct phononic band gaps in pistol-shrimp cuticle have been excluded pending claw-specific measurements.